



Run Ollama On Your Own Machine



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the capital
of France is



Introducing Today's Project!

In this project, I'm going to deploy Ollama on my local PC and chat with it without requiring any internet access. This will help me understand how to run LLMs locally. I'm interested in this because I want to upgrade my understanding of LLMs and limits of LLMs deployed locally.

Key tools and concepts

The key tools I used include Terminal, Ollama and different Ollama models. Key concepts I learnt include Local LLMs, model specifics, What is RAG, how to use Modelfile to create custom personas for my local LLM.

Challenges and wins

This project took me approximately 30 mins. The most challenging part was understanding the why behind what we were doing.



Installing Ollama for Local AI

In this step, I'm going to install Ollama on my mac using the terminal, confirm it is running and list the downloaded models.

A screenshot of a terminal window with a black background and white text. The text reads "ollama is running%" followed by a cursor.

Verifying the installation

I verified Ollama is running by using ``curl http://localhost:11434`` command in the terminal. The response I saw was ``Ollama is running``



Pulling My First AI Model

In this step, I'm going to check my current models, pull a model and then chat with my local AI. The model I'm using is qwen2.5:0.5b.

NAME	ID	SIZE	MODIFIED
qwen2.5:0.5b	a8b0c5157701	397 MB	13 seconds ago

Understanding model sizes

I pulled the model using ``ollama pull qwen2.5:0.5b`` command. qwen2.5 is a model developed by Alibaba. The 0.5b means that the model has 0.5 billion parameters.



Chatting with My Local AI

I chatted with my local AI by running ``ollama run qwen2.5:0.5b`` command and then asking it ``What is the capital of France``. It responded with ``The capital of France is Paris.``. This is different from cloud AI because the model did not use internet to search before responding.

```
>>> What is the capital of France?  
The capital of France is Paris.  
  
>>> /bye
```



Exploring Local AI Limitations

In this step, I'm going to discover what local AI models can and can't do. Local AI models can't know details for events happening in real-time, it also does not know details about me.

```
>>> What's my name?  
I am Owen, an artificial intelligence language model created by Alibaba Cloud. I don't have a physical form or a name as humans do. My purpose is to assist you with information and answer questions to the best of my abilities. If you have any other questions, feel free to ask!
```

Understanding RAG

The AI couldn't answer because it is only trained on general knowledge and doesn't have information on me. RAG stands for Retrieval-Augmented Generation and it is a technique that lets an LLM model bridge the gap between what it already knows from its training data and any private, custom, or real-time data.



Creating a Custom AI Persona

In this project extension, I created a persona called coding-tutor. It behaves differently because it was given specific system prompts to be friendly, use analogies and to break down concepts.

```
>>> What is a variable in programming?
Hello! Absolutely! 🍌 So glad you asked about variables – they're actually one of the very important concepts at the base of almost all coding languages!

Think of a variable as a labeled box where we can temporarily store something. A "box" isn't really useful if there's nothing inside it, right?

Let's look closely. Here's a breakdown:

What is a Variable First? 1)

* Imagine you have a basket of fruit. If you put "apple" in the basket, you've labeled the fruit in that box. If you take out one apple and replace it with an orange, you've actually altered the value inside the "box". That's basically what a variable does!

2) How They Work:
* A programmer assigns a name (like "age") to a specific location – the variable.
* We give it a type. (e.g., age is probably an integer—a number)

3) What You Can Store:
* Variables can store different things. It could be text ("Hello"), a number, even just dates. Think of them as containers with different contents

Let's make it simpler with examples!

Simple Example (Python):

```python
name = "Alice" #This declares a variable named 'name' and assigns the string "Alice" to it.
age = 30 #This declares a variable also named 'age' and assigned 30.

print("Hello," + name, ",", age) #Here we are concatenating strings in Python!
```

In this case: 1. We have variables 'name', 'age'. 2. The values for names can be assigned through the 'value' syntax.
What do you think? Do those examples make sense? Would you like me to explain them slightly further with code examples in more detail? Or would you like a different type of analogy to help us along?
```



Wrapping Up

I did this project today to learn how to run LLMs locally. Another skill I want to learn is how to use vector databases with local LLMs to give more larger context to the model.



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